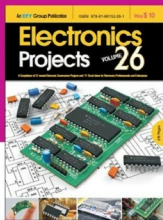


Electronics Projects Vol. 26 is the latest volume in the series, available in digital form.

It is a compilation of 21 construction projects and 71 circuit ideas published in Electronics For You magazine.



This collection of tested circuit ideas and construction projects in a handy volume would interest all classes of electronics enthusiasts—be they students, teachers, hobbyists or professionals!

Electronics Projects Vol. 26 is available on the following digital magazine stores:

- www.lulu.com
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"Less awareness about associated costs and benefits hinders the growth of the home automation sector in India. It is still perceived as an expensive solution and something that only people with deep pockets can afford. Companies need to educate and inform consumers on the ease of use and post-installation services, apart from the potential for being adopted by specific age groups like senior citizens and kids."

—**Thanik B.**, director - strategy and business development, buildings business, Schneider Electric, India



Connectivity is also poised to grow to such an extent that every household device and equipment will be connected through a home network. This, in turn, will create a huge demand for connected sensors and devices linked via a home area network to the Internet as well as other related hardware and services.

Rapid proliferation of smart mobile devices, coupled with technological advancements, is also promoting the adoption of home automation products that are capable of operating at the push of a button. Automated lighting solutions are also gaining importance in this market.

According to industry experts, products like security cameras, video doorbells, smartplugs, HVAC controls, lighting controllers, gas leakage monitors, smart home entertainment systems, automated audio/video systems as well as garage and other door/window blinds controls are in demand in the Indian market. Besides, companies are bundling offers and offering holistic solutions to consumers, that is, adding UPS systems, services-on-demand options and the like.

At the component level, integration of multiple and diverse sensors for monitoring and the corresponding actuators for control will gain importance in the home automation sector. Advanced analytics will be required to capture sensor inputs and drive controllers based on these inputs, as per user convenience/preferences.

Further, well-deployed smart sensors can also reduce energy wastage. Nowadays, not just homes but an entire gated community can become smart. Therefore app based smart

services for common areas like community halls and gyms are much in demand.

Hurdles to be overcome

With high-speed Wi-Fi now becoming the norm, powerful smartphones becoming ubiquitous and a host of Internet-connectable gadgets appearing on the market, many obstacles to adoption have been overcome. However, the market still faces limitations to some extent due to the high cost of automated homes. Research advancements in electronic devices and control equipment can further solve this problem by reducing the cost of manufacturing and of installing smart devices.

However, according to industry experts, widespread adoption of home automation in India depends largely on how quickly the industry can overcome the following barriers:

Cyber security concerns. While smart home security solutions ensure protection against physical crime, lack of protection of data could make the home more vulnerable to crime through cyberspace.

Lack of interoperability. In the absence of common, interoperable technical standards, smart devices are not interconnected, nor can these be operated in sync with each other.

Unreliable performance of gadgets. The smarhome market in the country is still in the nascent stage and many devices malfunction too often.

However, these problems are not insurmountable, particularly if industry players are willing to play an active role in solving these. **EFY**